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VisualParadigm
For non-commercial use

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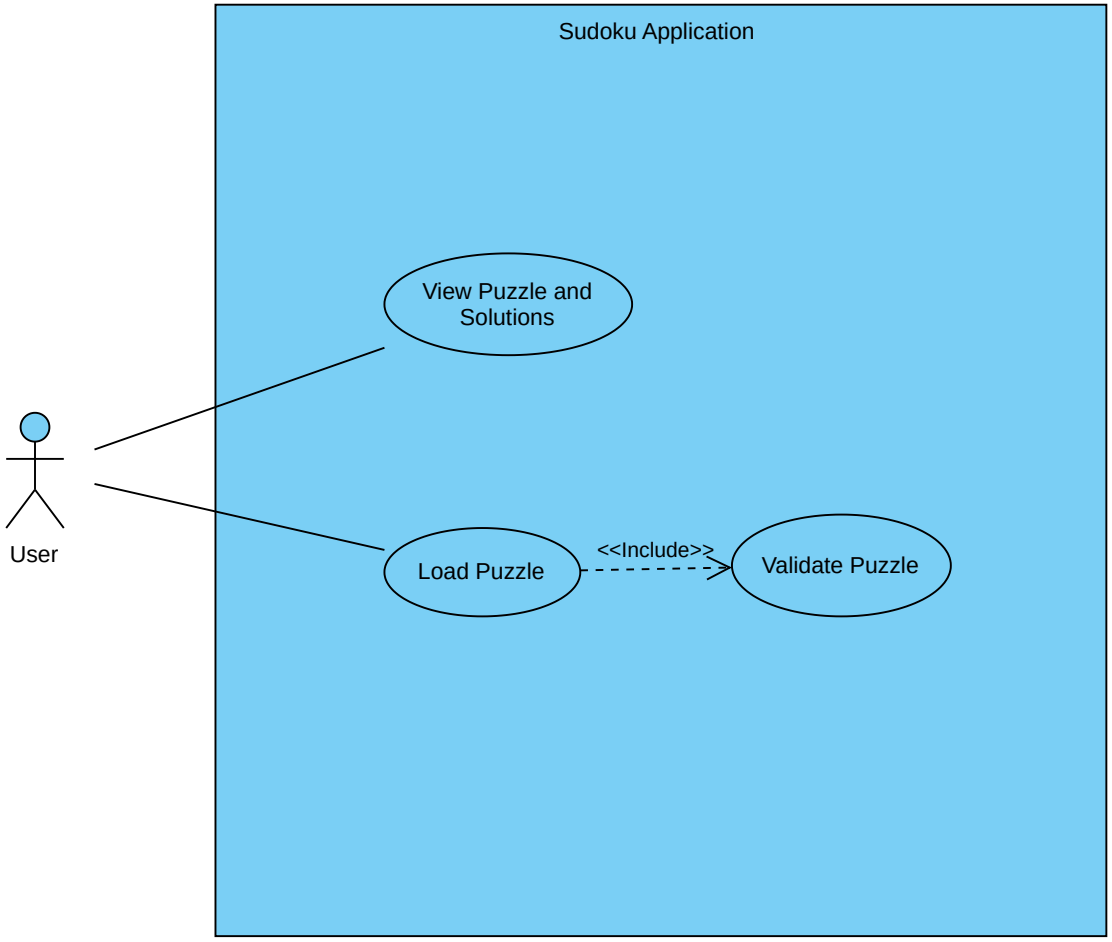

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a. Identify/discover a use case
   i. Load Puzzle
b. Provide a brief description for the use case
   i. A program loads user inputted puzzle and checks if it is a valid input
c. Provide a step-by-step flow (outline) of the events for the use case
   i. User inputs puzzle into program
   ii. Program checks if input is valid then loads up necessary information to solve
d. Provide a detailed description for the use case
   i. Program takes in users 9x9 sudoku puzzle and checks if it is is a 9x9 grid, file is valid, no invalid characters.

a. Identify/discover a use case
   i. View Puzzle and solutions
b. Provide a brief description for the use case
   i. User opens text file to view validated puzzle
c. Provide a step-by-step flow (outline) of the events for the use case
   i. User opens file modified by program
   ii. User views original puzzle and its solution(s) if there is one
d. Provide a detailed description for the use case
   i. User opens the text file modified by the program. They can view what file it is named, the original puzzle itself, all possible solutions, number of solutions or text that states there is no solution

a.
Identify/discover a use case
i. Validate Puzzle
b.
Provide a brief description for the use case
i.
ii.
c.
Provide a step-by-step flow (outline) of the events for the use case
i. A program takes in loaded puzzle
ii. Attempts to solve it by depth first search
iii. once a solution is found save it
iv. continues until every option has been attempted
v. outputs original puzzle, the solutions, number of solutions
vi. if no solutions outputs no solution
d.
Provide a detailed description for the use case
i. Program takes in loaded puzzle then attempts to solve it. It does this by using a depth first search approach with backtracking. It continually replaces empty slots with valid inputs and checks if the puzzle is valid if not it backtracks to first invalid point and changes it and repeats this until it has exhausted every possible option

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